

Sulforaphane protects human chondrocytes against cell death induced by various stimuli.



Chondrocyte cell death can contribute to cartilage degeneration in articular diseases, such as osteoarthritis (OA). Sulforaphane (SFN), a natural compound derived from cruciferous aliment, is well known as an anti-carcinogen, but according to recent evidence it also shows cytoprotective effects on a variety of non-tumoral cells. Therefore we have tested the ability of SFN to protect chondrocytes from cell death in vitro. Treatment of growing monolayer cultures of human C-28/I2 chondrocytes with SFN in the low micro-molecular range for a few days, reduced cell growth without affecting cell survival or inducing apoptosis. However it decreased cell death in C-28/I2 chondrocytes exposed to stimuli previously reported to promptly trigger apoptosis, that is, the cytokine tumor necrosis factor- α (TNF) plus cycloheximide (CHX) or the polyamine analogue N(1),N(11)-diethylnorspermine (DENSPM) plus CHX. In particular pre-treatment with SFN reduced effector and initiator caspase activities and the associated activation of JNK kinases. SFN exerted a cytoprotective action even versus H₂O₂, which differently from the previous stimuli induced cell death without producing an evident caspase activation. SFN pre-treatment also prevented caspase activation in three-dimensional micromass cultures of OA chondrocytes stimulated with growth-related oncogene α (GRO α), a pro-apoptotic chemokine. The suppression of caspase activation in micromasses appeared to be related to the inhibition of p38 MAPK phosphorylation. In conclusion, the present work shows that low micro-molecular SFN concentrations exert pro-survival and anti-apoptotic actions and influence signaling pathways in a variety of experimental conditions employing chondrocyte cell lines and OA chondrocytes treated with a range of death stimuli. Copyright © 2010 Wiley-Liss, Inc.